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Introduction new combination of zero-order hold and first-order hold

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***Abstract*— This work introduces a new combination of holders with application on digital control systems .This holder is a combination of zero-order hold (ZOH) and first-order hold (FOH) that has the capability of both holders. It is obvious that, At low frequencies, FOH is a better approximation to an ideal low-pass filter than ZOH, but at high frequencies, ZOH is better than FOH. Because of this fact, we introduced a combination of ZOH and FOH, we used of a frequency switch, that it has been set with input frequency, that it switches by input frequency between ZOH and FOH, therefore it has the same behavior as FOH at low frequencies and the same behavior as ZOH at high frequencies, and hence we suggested a combination of ZOH and FOH that it works more accurate at two conditions, low and high frequencies. Also simulation results are studied to show the effectiveness of the proposed holder and better performance results in comparison with ZOH and FOH.**

***Index Terms*— Combination, frequency switch, first-order hold, zero-order hold.**

I. INTRODUCTION

The dynamics of the systems are naturally continuous, while control inputs are usually applied at discrete time instants. This form of the sampled-data systems arises in various applications and system formulations. Thus, the control of sampled-data systems is an important practical problem, and its controller design method is necessary in many applications. Many works for the sampled-data systems have ever been done and the significant results have appeared in the literature (for example, Chen and Francis [1], Sun et al. [11], Farzad Hashemzadeh et al. [2],and references therein).Fridman et al. [3] introduced and developed a delay system approach to sampled-data stabilization of linear systems. The continuous-time linear system with sampled-data control input results in the closed-loop system with time-varying state-delays. Sufficient stability conditions for such a linear state delayed system were obtained and a design method of a stabilizing sampled data state feedback controller was proposed. Further development has been made in Fridman and Shaked [4], Suplin et al. [12] and Yoneyama [8] where H∞ sampled-data control and robust stabilization of time-delay systems were concerned. However, these results only consider the zero-order hold control.The zero-order hold control is easy to be implemented but may not be an appropriate control law on the sampling intervals.

In this work, a new holder to be applied to digital control systems is introduced. we named it mixed zero-first order hold (ZFOH). ZFOH is a combination of ZOH and FOH that has the capabilities of both holders and has a Better response than ZOH and FOH. At low frequencies, FOH is a better approximation to an ideal low-pass filter than ZOH, but at high frequencies, ZOH is better than FOH. Because of the fact that ZFOH is a combination of ZOH and FOH, it has the same behavior as FOH at low frequencies and the same behavior as ZOH at high frequencies, and hence by dividing frequencies in to two sections, high and low frequencies, we can have a better frequency response than ZOH and FOH. By the use of a frequency switch[13], that it has been set with input frequency, we produced an optimum holder, when the frequency of input wave is high, the output is from ZOH, and when the frequency of input wave is low, the output is from FOH, that we commonly named it ZFOH.

In Section 2 the equations of holders is presented. The Simulation Results of frequency response, ZOH, FOH and ZFOH is given in Section 3 and was compared. in section 4 A numerical example is given to illustrate the advantages of the ZFOH, finally in section 5 conclusion is presented

II. EQUATIONS

A. ZOH equation

The ZOH is the most common holder used as digital to analog converter (DAC) in industry, that it introduce from first term of taylor series with this form[5,6]:

$$G_h = \frac{1-e^{-Ts}}{s} \quad (1)$$

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B. FOH equation

We have transfer function of the first-order hold that it made from first and second terms of talor series like as below :

$$G_{h1}(s) = \left(\frac{1-e^{-Ts}}{s}\right)^2 \frac{Ts+1}{T} \qquad (2)$$

C. ZFOH equations

In this equations, we combined ZOH and FOH, for high and low frequencies like as below:

$$\begin{cases} G_h = \dfrac{1-e^{-Ts}}{s} & For\ High\ Frequenc \\ G_{h1}(s) = \left(\dfrac{1-e^{-Ts}}{s}\right)^2 \dfrac{Ts+1}{T} & For\ Low\ Frequen \end{cases} \qquad (3)$$

In this system when input is with high frequency the output will be switch on ZOH and when input is with low frequency the output will be switch on FOH, hence we will have a optimum system that works with high accuracy in to two sections.

III. SIMULATION RESULTS

In this part, simulation results of the comparison among the proposed holders, ZOH and FOH, are given.
We applied a sinusoidal wave at high and low frequencies and draw that:

A. At the low frequency for ZOH we have

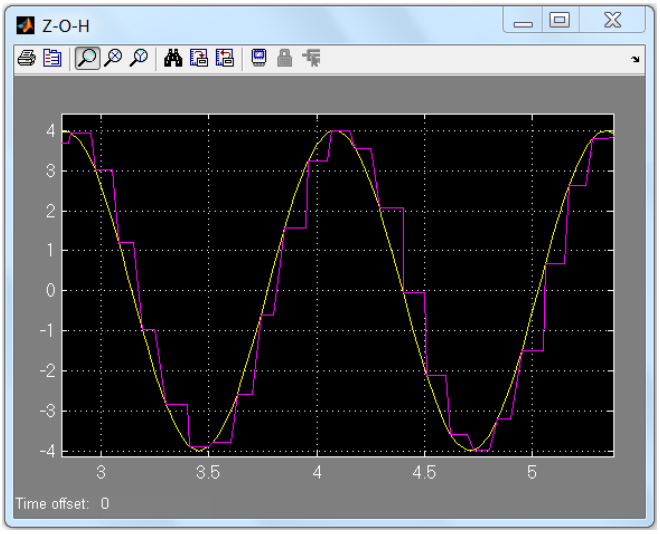


Fig. 1 Behavior of ZOH at the low Frequency

B. At the high frequency for ZOH we have

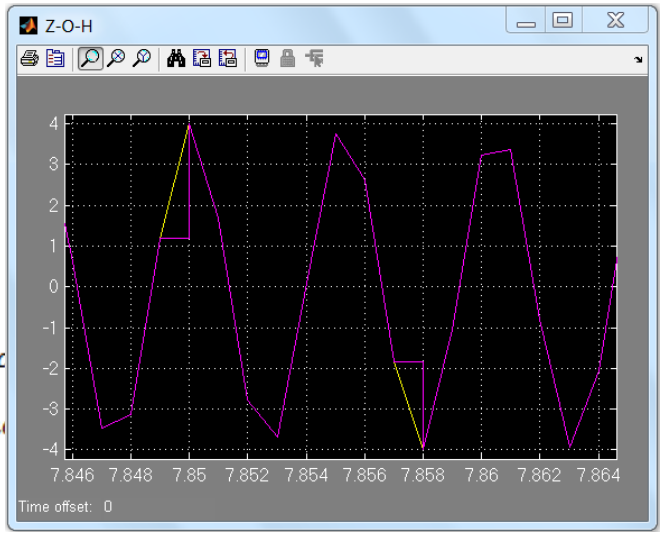


Fig 2 Behavior of ZOH at the high Frequency

C. At the low frequency for FOH we have

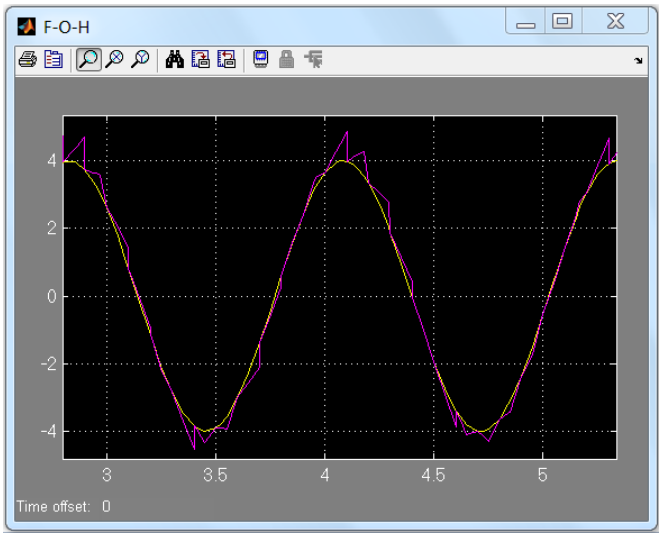


Fig. 3 Behavior of FOH at the low Frequency

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D. At the high frequency for FOH we have

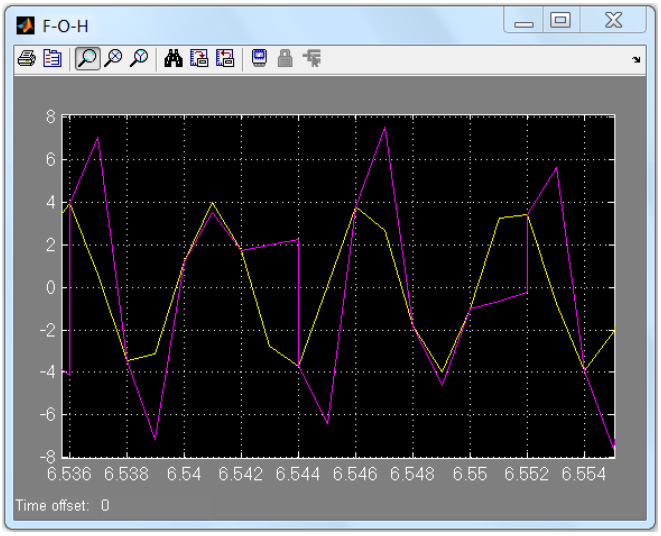


Fig. 4 Behavior of FOH at the high Frequency

IV. NUMERICAL EXAMPLE

In this part to illustrate better performance of our holder, present an example.
Consider the following function :

$$F(s) = \frac{2}{s^2+20s+100} \quad (4)$$

The frequency response of function at low and high frequency presented at the continue :

At the low frequency

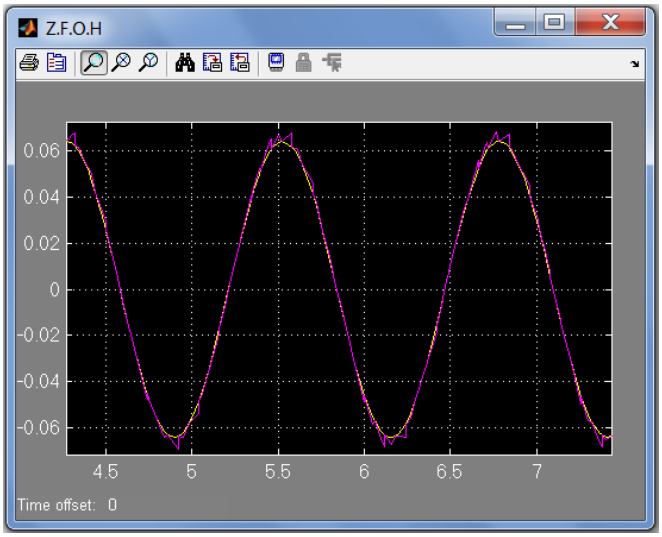


Fig. 5 Behavior of ZFOH at the low Frequency

At the high frequency

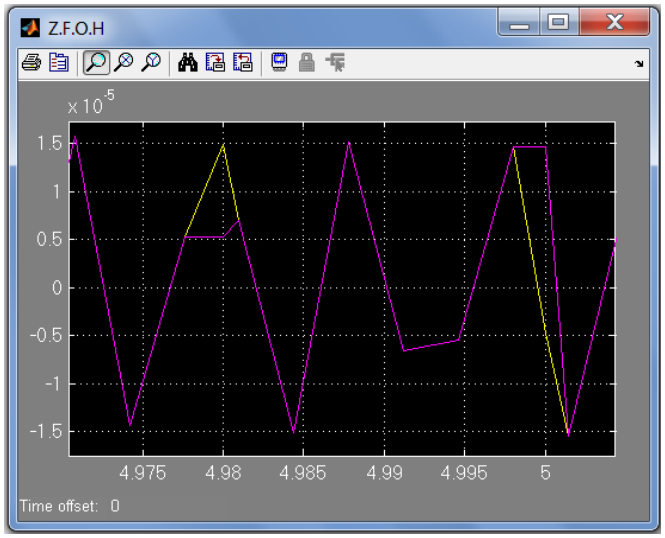


Fig. 6 Behavior of ZFOH at the high Frequency

We see that in ZFOH system, we have an optimum output, at high and low frequency and maximum similarly to applied wave.

V. CONCLUSION

In this paper we introduced a new holder for digital control systems that has a better performance in terms of the frequency response than ZOH, FOH, and has better frequency response. by use of a frequency switch, that it has been set with input frequency, that it switches by input frequency between ZOH and FOH, therefore it has the same behavior as FOH at low frequencies and the same behavior as ZOH at high frequencies, and hence we suggested ZFOH that it is a combination of ZOH and FOH, it works more accurate at two conditions, low and high frequencies. This new holder has the capability both of ZOH and FOH. Finally, we gave an example and showed our results, makes a better holder than the ZOH and FOH.

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